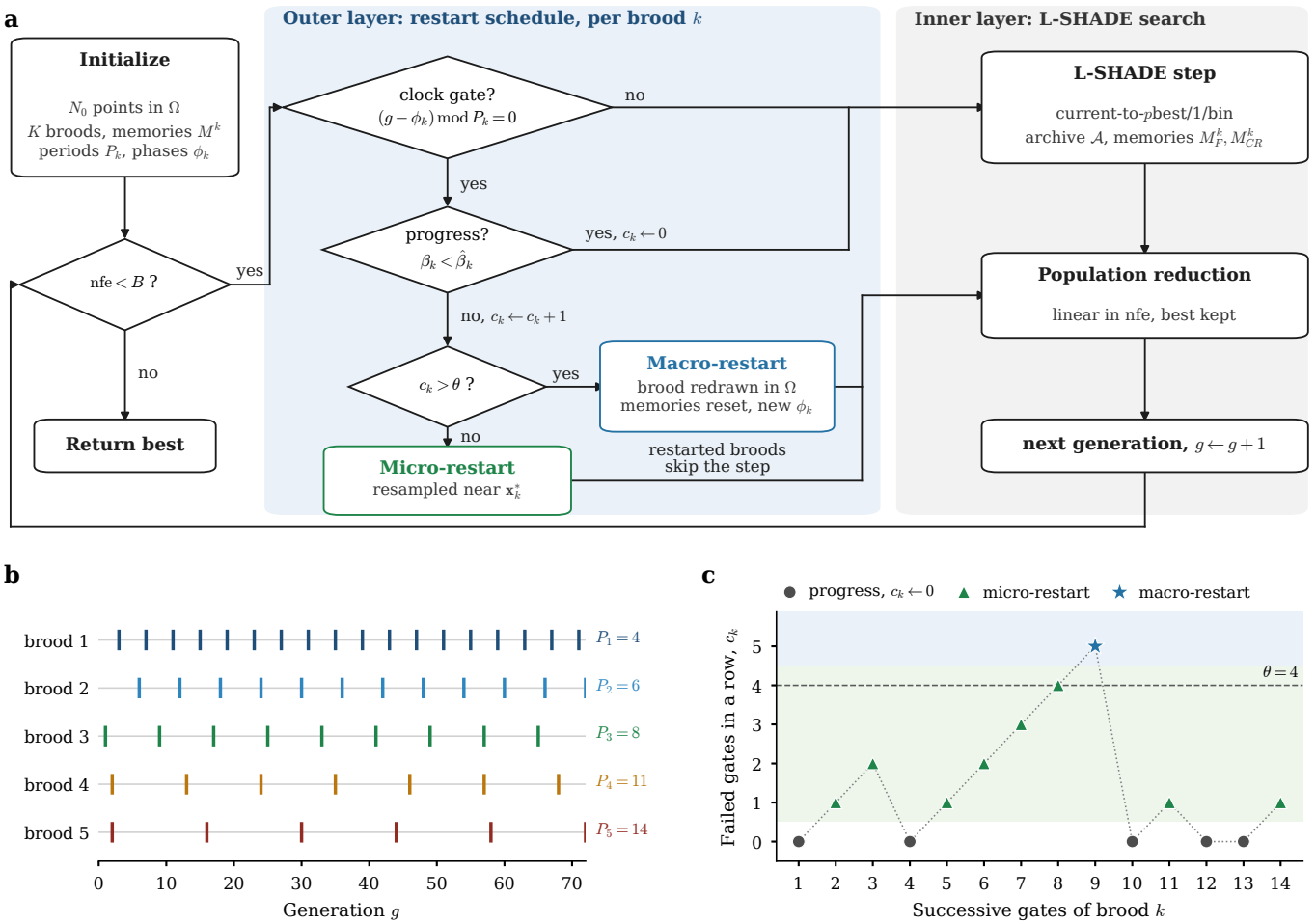
a
Outer layer: restart schedule, per brood k
Inner layer: L-SHADE search
Initialize
N_0 points in Ω
K broods, memories M^k
periods P_k, phases ϕ_k
clock gate?
$(g-\phi_k) \bmod P_k = 0$
no
yes
progress?
$\beta_k < \hat{\beta}_k$
yes, $c_k \leftarrow 0$
no, $c_k \leftarrow c_k + 1$
$c_k > \theta$?
yes
no
Macro-restart
brood redrawn in Ω
memories reset, new ϕ_k
Micro-restart
resampled near $\mathbf{x}^*_k$
restarted broods
skip the step
nfe $< B$?
yes
no
Return best
L-SHADE step
current-to-$p$best/1/bin
archive $\mathcal{A}$, memories M^k_F, M^k_{CR}
Population reduction
linear in nfe, best kept
next generation, $g \leftarrow g+1$
b
brood 1
brood 2
brood 3
brood 4
brood 5
$P_1 = 4$
$P_2 = 6$
$P_3 = 8$
$P_4 = 11$
$P_5 = 14$
0
10
20
30
40
50
60
70
Generation g
c
progress, $c_k \leftarrow 0$
micro-restart
macro-restart
Failed gates in a row, c_k
$\theta = 4$
0
1
2
3
4
5
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Successive gates of brood k